Report LINFO1361: Assignment 2

Group N°... (Moodle and Inginious)

Student1: .....

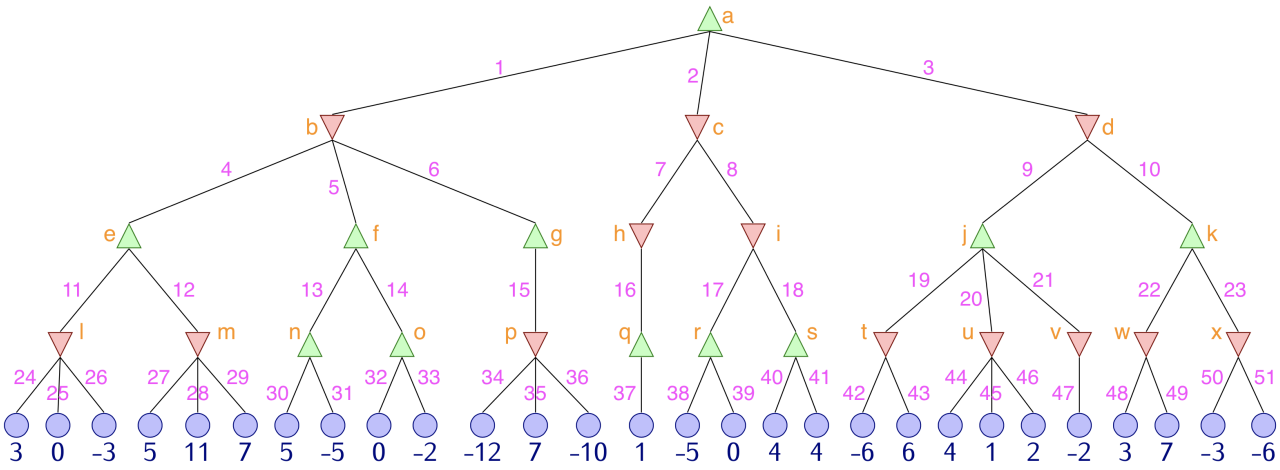
Student2: .....

April 6, 2024

Answer to the questions by adding text in the boxes. You may answer in either **French or English**. Do not modify anything else in the template. The size of the boxes indicate the place you **can** use, but you **need not** use all of it (it is not an indication of any expected length of answer). **Be as concise as possible! A good short answer is better than a lot of nonsense!**

1 Exercises (5 pts)

The following figure assigns a unique letter to each node, and a unique number to each branch. Use it to answer the following questions.



1. Perform the MiniMax algorithm on the following tree, i.e. put a value to each node. What move should the root player do? **(1 pt)**

   Assign a numerical value to each node, and indicate the move (i.e. 1, 2, or 3) to perform:

a:	**f:**	**k:**	**p:**	**u:**
b:	**g:**	**l:**	**q:**	**v:**
c:	**h:**	**m:**	**r:**	**w:**
d:	**i:**	**n:**	**s:**	**x:**
e:	**j:**	**o:**	**t:**	**Move:**

2. Perform the Alpha-Beta algorithm on the same tree. At each non terminal node, put the successive values of α and β. Cross out the arcs reaching non visited nodes. Assume a left-to-right node expansion. **(1 pt)**

   Indicate the successive α and β values of each node in the table below. Separate successive values by a comma (,). Indicate at the bottom the identifiers of the branches that are cut (in increasing order, separated by a comma) (indicate only the branches where the cuts happen, i.e. don't *indicate the branches that are below a cut).*

Node	α values	β values
a		
b		
c		
d		
e		
f		
g		
h		
i		
j		
k		
l		
m		

Node	α values	β values
n		
o		
p		
q		
r		
s		
t		
u		
v		
w		
x		

Cuts:

3. Do the same, assuming a right-to-left node expansion instead. **(1 pt)**

Node	α values	β values
a		
b		
c		
d		
e		
f		
g		
h		
i		
j		
k		
l		
m		

Node	α values	β values
n		
o		
p		
q		
r		
s		
t		
u		
v		
w		
x		

Cuts:

4. Is there a node ordering that can lead to a more optimal pruning of the tree (in the sense where the algorithm prunes more branches than in the two other considered cases)? If no, explain why. If yes, give a new node ordering and the resulting new pruning. **(1 pt)**

   Insert an image below containing the reordered tree, with successive α/β values indicated next to each node, and where the branches that are cut by the algorithm are crossed out. This may either be an edited version of `minimax_empty.png` *(using paint, gimp, etc.), a photograph of a drawing you made by hand, etc. In any case, the image must be* ***clear*** *in order to be graded.*

5. How does Alpha-Beta need to be modified for games with more than two players? **(1 pt)**

2 Shobu (35 pts)

2.1 Alpha-Beta agent (4 points, to be submitted on Inginious)

2.2 Monte-Carlo Tree Search agent (5 points, to be submitted on Inginious)

2.3 Warm-up questions (3 points)

1. What is the branching factor at the start of the game? What is the mean empirical branching factor? (The branching factor is considered here as the number of possible moves that a player can do from a given state). **(1 point)**

2. What would be one (of the many) drawbacks of the simple heuristic that is imposed for the basic alpha-beta agent above? **(1 point)**

3. Considering the Monte-Carlo Tree Search algorithm, what is the hypothesis supporting the use of random simulation to estimate the win rate? Is this hypothesis always valid? **(1 point)**

2.4 Description of your agent (8 points)

Describe in the boxes below what you have implemented for your contest agent. You can also mention things that you tried and did not work but focus on what you have submitted in the end.





2.5 Comparison of agents (5 points)

Describe here the comparison of the different agents: random, Alpha-Beta, MCTS and your agent (and others if you want!). Remember that it should be a statistical comparison. Describe how you compare the agents. Draw some observations and conclusions based on the results you have obtained.





2.6 Contest (10 points, to be submitted on Inginious)